



Genesis Biotech Inc.

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bFGF (Basic fibroblast growth factor)

Catalog No. PG-40003

Scale: 100 µg

Protein species: Human

Bioactivity (ED50): 0.1-0.3 ng/ml

Applications tested: cell culture

Expression Host: *E. coli* **Type:** purified recombinant protein

Supply form: lyophilized protein

Target description

Basic fibroblast growth factor, otherwise known as bFGF or FGF-2, was the second discovered member of the FGF factor family (a family of structurally related polypeptides, characterized by high affinity to heparin and heparan sulfate), and is the most widely studied. bFGF binds with high affinity to specific tyrosine kinase receptors and with lower affinity to cellular heparan sulfates. It has anti-apoptotic effects and is a potent mitogen for cells of mesenchymal, neural, and epithelial origin. It also is a potent angiogenic agent, both in vitro and in vivo, due to its proliferative effects on vascular endothelial cells. Due to its unique properties, bFGF can be stored in the basement membrane and then released by administration of heparin or by enzymatic matrix degradation. The bFGF has strongly been implicated to play roles in a large number of normal and pathologic settings, such as the development of the skeletal system, wound healing, and angiogenesis.

Potential applications

1. An important regulator of endothelial cell (EC) proliferation, migration, and protease production.
2. Wound healing.
3. Angiogenic agent
4. Protection neurons of various structures from excitotoxic damage
5. Anti-apoptotic effects

Comments

bFGF(Recombinant human basic fibroblast growth factor): Molecular Mass: 16.5kDa, composed of 147 aa. Human bFGF is recombinant DNA produced and >95% pure via SDS-page and HPLC.

Related Products

1. IGF-1(insulin-like growth factor-1)(PG-40001)
2. PDGF-B (platelet-derived growth factor B-chain) (PG-40002)
3. Fibronectin (PG-40004)
4. TNF α (Tumor necrosis factor -alpha) (PG-40005)

Antibiotic residue

No activity of ampicillin or other antibiotic residue detected.

Bacterial endotoxin content

Bacterial endotoxin content is not higher than 0.1ng/ug (1EU/ug)

Biological Activity

The ED50 is defined as the effective concentration of growth factor (dose)that elicits a 50% increase in cell growth in a cell based bioassay. ED50: 0.1-0.3 ng/ml

Solubility

It is recommended to reconstitute the lyophilized bFGF in sterile H₂O not less than 100µg/ml, which can then be further diluted to other aqueous solutions.

Storage

For short-term use store at 4 °C. For long-term storage keep at -20°C.

References

1. Laham R, Simons M. Therapeutic angiogenesis in myocardial ischemia. *Angiogenesis and Cardiovascular Disease*. 289-320, 1999.
2. Folkman J: Angiogenic therapy of the human heart. *Circulation*. 97:628-629, 1998.
3. Gospodarowicz D: Expression and control of vascular endothelial cells: proliferation and differentiation by fibroblast growth factors. *J Invest Dermatol*. 93:39S-47S, 1989.

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